ECONOMETRICS I
Practice Problem Set 1

The textbook is Jeffrey M. Wooldridge Introductory Econometrics: A Modern Approach, 6th ed., 2016, Cengage Learning

PROBLEMS

1. The joint probability distribution of random variables X and U is given in the table below.

		U		
		-1	0	1
X	1	0.35	0.1	0.1
	2	0.1	0.15	0.2

   Table entries are the joint probabilities (e.g., $f(U = -1, X = 1) = 0.35$). Using this table compute $E[U|X = 1]$.(Hint: first compute conditional probabilities using $f(u|x) = f(u, x)/f(x)$, where $f(x)$ is the marginal distribution function for X, then apply the conditional expectation formula.)

2. The following model aims to explain the causal relationship between education spending and success at the university entrance exams:

$$TestScore = \beta_0 + \beta_1 Expenditure + u$$

   where *TestScore* is the score from university entrance exam, and *Expenditure* is total education expenditures.

   (a) Can we say that the causal relationship between *Expenditure* and *TestScore* is established in this model? Why or why not? Explain.

   (b) Suppose that $u = \epsilon\sqrt{Expenditure}$ where ϵ is a random variable independent from *Expenditure* with $E(\epsilon) = 0$ and $Var(\epsilon) = \sigma_\epsilon^2$. Find $Var(u|Expenditure)$

3. Is $y = \exp(\beta_0)x_1^{\beta_1}x_2^{\beta_2}\exp(u)$ linear in parameters? If not, write this model as linear in parameters.

4. The data set contains data on births to women in the United States and the data comes from random sampling.Two variables of interest are the dependent variable, infant birth weight in ounces (bwght), and an explanatory variable, average number of cigarettes the mother smoked per day during pregnancy (cigs). The following simple regression was estimated using data on (sample size)n=1388 births:

$$\widehat{bwght} = 119.77 - 0.514 cigs$$

   (a) What is the predicted birth weight when *cigs =0?* What about when *cigs=20* (one pack per day)? Comment on the difference.

   (b) Does this simple regression necessarily capture a causal relationship between the child 's birth weight and the mother 's smoking habits? Explain.

5. Consider the savings function

$$sav = \beta_0 + \beta_1 inc + u$$

$$u = \sqrt{inc}.e,$$

where e is a random variable with E(e)= 0 and Var$(e) = \sigma_e^2$. Assume that e is independent of *inc*.

(a) Show that E$(u|inc)$= 0, so that the key zero conditional mean assumption (Assumption SLR.4) is satisfied. [Hint: If e is independent of *inc*, then E$(e|inc)$= E(e).]

(b) Show that Var$(u|inc)$= $\sigma_e^2 inc$, so that the homoskedasticity Assumption SLR.5 is violated. In particular, the variance of *sav* increases with inc. [Hint: Var$(e|inc)$= Var(e), if e and *inc* are independent.]

(c) Provide a discussion that supports the assumption that the variance of savings increases with family income.

6. Suppose that the following models are estimated in order to determine the relationship between y and, x_1, x_2. Using these results answer the following questions.

			$\hat{\beta}_0$		$\hat{\beta}_1$		$\hat{\beta}_2$	
Model A:	$\widehat{y}$	=	4.12	+	0.80 x_1	+	0.16 x_2	$R_A^2 = 0.22$
Model B:	$\widehat{\log(y)}$	=	0.34	+	0.17 x_1	+	0.001 x_2	$R_B^2 = 0.25$
Model C:	$\widehat{\log(y)}$	=	-2.01	$-$	0.82 $\log(x_1)$	+	0.02 x_2	$R_C^2 = 0.32$
Model D:	$\widehat{y}$	=	-3.4	+	43.7 $\log(x_1)$	$-$	0.020 x_2	$R_D^2 = 0.28$

(a) According to Model A: If $\Delta x_1 = 6$, then, ceteris paribus, $\Delta\hat{y}$ =.....

(b) In Model B, interpret $\hat{\beta}_1$.

(c) In Model C, interpret $\hat{\beta}_1$.

(d) In Model D, in order for the increase in $\widehat{y}$ to be 1 unit we must have Δx_2=..... .

(e) In Model D, interpret $\hat{\beta}_1$.

(f) In Model A, interpret the coefficent of determination.

Review Questions:
(These will not be solved. Read the classnotes and the textbook)

7. Define observational and experimental data and discuss the differences.

8. List and briefly explain the steps involved in an econometric analysis. What is the difference between an economic model and an econometric model?

9. Briefly discuss the finite sample properties of OLS estimators (unbiasedness and efficiency). What assumptions are needed for unbiasedness in the context of simple regression model?